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Review

Autoimmunity in inflammatory bowel disease: a holobiont perspective

Henry Taylor[1], Holm H Uhlig[2,3,4] and Fiona Powrie[1,4]

Adaptive immunity towards self-antigens (autoimmunity) and intestinal commensal microbiota is a key feature of inflammatory bowel disease (IBD). Considering mucosal adaptive immunity from a holobiont perspective, where the host and its microbiome form a single physiological unit, emphasises the challenge of avoiding damaging responses to self-antigen and symbiotic microbial communities in the gut while protecting against potential pathogens. Intestinal tolerance mechanisms prevent maladaptive T and B cell responses to microbial, environmental, and self-antigens, which drive inflammation. We discuss the spectrum of antimicrobial and autoantibody responses and highlight mechanisms by which common IBD-associated adaptive immune responses contribute to disease.

Addresses
[1] Kennedy Institute of Rheumatology, University of Oxford, Oxford, UK
[2] Centre for Human Genetics, University of Oxford, Oxford, UK
[3] Translational Gastroenterology Liver Unit, University of Oxford, Oxford, UK
[4] NIHR Biomedical Research Centre, University of Oxford, Oxford, UK

Corresponding author: Taylor, Henry (henry.taylor@kennedy.ox.ac.uk)

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Introduction

Autoimmunity in inflammatory bowel disease (IBD) has been recognised for over 60 years since the first IBD-associated autoantibodies were detected against colon extract in 1959 [1], but views on its contribution to the pathogenesis of IBD have evolved over time. Today IBD is considered a complex immune-mediated inflammatory disease characterised by defective mucosal barrier function and excess pro-inflammatory innate and adaptive immune responses to a dysbiotic microbiome [2,3].

We consider mucosal adaptive immunity from a holobiont perspective, in which the host and its highly associated co-developing microbial communities are viewed as a single physiological unit. This shift in perspective emphasises that maintenance of tolerance to both self-antigen and the symbiotic microbiota is critical for mucosal homeostasis. It positions loss of tolerance as an important checkpoint in the development of IBD.

We review the roles of the mucosal adaptive immune system in health and highlight mechanisms that provide host-protective immunity and maintain tolerance. We delineate how innate (auto)inflammatory processes and specific T and B cell defects contribute to loss of tolerance. Reviewing evidence from inborn errors of immunity (IEI) and mouse models, we show that autoimmune and inappropriate antimicrobial adaptive immune responses arise from defects in overlapping mucosal tolerance and immunoregulatory systems. We consider autoimmunity as part of a larger group of IBD-associated mucosal adaptive responses. We discuss how these responses can be bystander reactions with utility as biomarkers or can contribute to disease pathogenesis and present new therapeutic targets.

Section 1: Immune homeostasis in the holobiont

The intestinal microbiome is a complex community of organisms that exists in symbiosis with its host. Microbiota perform essential digestive, metabolic, and immunomodulatory functions in health, and disturbance of the microbiota can contribute to disease [4]. The close, mutually dependent relationship between host and its microbiome has led to the two being considered together as a single physiological unit, a holobiont [5].

This perspective highlights the challenge posed to the immune system at mucosal sites. Symbiotic microbial communities are part of the 'holobiont-self' and must be maintained, yet the intestinal lumen is not an immunologically privileged site. The mucosa is an active interface, where the immune system must support and contain beneficial luminal microbiota while protecting against pathogenic bacteria, viruses, fungi, and parasites.

Maintaining homeostasis at this mucosal interface is complex. The epithelial barrier and innate immune system play an important role [2]. Mucins secreted by specialised epithelial cells form an important defensive

barrier [6], while pathogen recognition responses by enterocytes and innate immune cells control microbial ingress and influence immune responses via cytokine and interferon production [7]. In this review of autoimmunity in IBD, we focus on antigen-reactive responses of the adaptive immune system. Adaptive immunity influences the composition of microbial communities, supports barrier function, and employs immunoregulatory effector functions to promote tolerance.

The mucosal adaptive immune system — host-protective immunity and immune homeostasis

In the intestinal lumen and mucus layer, microbial communities are influenced by secretory IgA produced by intestinal plasma cells in the lamina propria (LP). IgA affects luminal microbiota through immune exclusion, neutralisation of virulence factors, agglutination, and enchainment [8]. This limits bacterial translocation [9] and creates ecological niches for symbiotic bacteria [10] and fungi [11] by altering colonisation topography and targeting competing pathobionts.

In the outermost layer of the gastrointestinal mucosa, intraepithelial lymphocytes (IELs) contribute to host protection and immune homeostasis. IELs comprise tissue-resident memory CD4+ and CD8+ T cells (T_{RM} cells) and unconventional T cells. Multiple regionally distinct T_{RM} populations provide rapid, antigen-specific responses against intestinal pathogens [12]. Functional specialisation of these responses can be determined by the tissue niche, for example, small intestinal CD8+ T_{RM} cells in the upper villus provide effector responses, while those in the lower villus contribute to a memory response [13]. Unconventional T cells have atypical T cell receptors (TCRs) with reduced diversity, enabling rapid responses to a limited pool of antigens. Mucosa-associated invariant T (MAIT) and natural killer T (NKT) cells possess αβ-TCRs, which recognise microbial antigens from riboflavin biosynthesis presented on MR1, and lipids presented on CD1d, respectively [14]. The γδ-TCR directly recognises microbial and host-derived ligands without major histocompatibility complex (MHC) antigen presentation. γδ-T cells alter the small intestine microbiome through faecal micro-RNA secretion [15] and present antigen to CD4+ T cells influencing their polarisation [16]. Single-cell analysis of the intraepithelial compartment reveals extensive functional heterogeneity, encompassing direct cytotoxicity against infected cells and pathogens, production of pro-inflammatory and barrier-supporting cytokines, antigen presentation, and tissue repair [12,17].

In the LP, CD4+ T cells also support mucosal barrier function. Th17 cells produce IL17A, IL17F, and IL22, which stimulate intestinal epithelial cells (IECs) to maintain tight junctions and produce antimicrobial peptides [18]. Pro-inflammatory cytokines, such as interferon gamma (IFNy), tumour necrosis factor alpha (TNFα), and granulocyte-macrophage colony-stimulating factor (GM-CSF), produced by Th1 and some Th17 cells, recruit and activate innate immune cells to clear invasive organisms [19]. In addition to TCR-mediated Th1/Th17 effector functions, an abundant population of microbiota-reactive CD4+ $CD161^{High}$ T cells (CD4/CD8 double negative in mice) contribute to immune homeostasis through TCR-independent responses to cytokine stimulation and by TCR-dependent tissue repair activity [20].

IgA and an effective mucosal barrier contain microbiota and limit antigen exposure. This combines with specific immunoregulatory tolerance mechanisms to maintain homeostasis.

Section 2: Tolerance and immune regulation in the intestinal mucosa

FOXP3+ Tregs have a key role in maintaining tolerance to the microbiota and self-antigen [21]. Tregs can differentiate in the thymus (tTregs) in response to high-affinity interaction with self-antigen on thymic antigen presenting cells (APCs) (e.g. *AIRE*-expressing medullary thymic-epithelial cells), so have a TCR repertoire enriched for self-antigen. Highly autoreactive thymic T cells can alternatively undergo clonal deletion, reducing the pool of self-reactive naïve effector T cells in the periphery [19]. Microbial antigen can be trafficked to the thymus by specialised APCs, which support education of microbiota-responsive effector T cells; it is not yet known whether they contribute to generation of microbiota-responsive tTregs [22].

Tregs are also generated in the periphery (pTregs) in response to specific microbiota such as *Clostridia* strains and *Helicobacter hepaticus* [19,23]. *Helicobacter*-specific pTreg differentiation is supported by specialised RORγt + APCs in mesenteric lymph nodes (mLN), which express the transforming growth factor beta (TGF-β) activating integrin $\alpha_v\beta_8$ [24]. In intestinal lymphoid tissue, circulating tTregs and locally differentiated pTregs upregulate GPR15, CCR9, and α4β7 in response to the intestinal milieu, enabling migration to the LP [21]. Intestinal Tregs then form a distinct population, with prolonged tissue residency, preferential gut homing, and unique TCR repertoires [25].

Within the intestine, pTregs and tTregs have common functionality and are not separated by unbiased clustering of single-cell RNA-sequencing data [23,26]. They are maintained and proliferate in an LP micro-niche, supported by interactions with CD206+ macrophages, IL1B+ macrophages, and CD103+ dendritic cells [26]. They have multiple tolerogenic effector functions, including production of immunoregulatory cytokines (TGF-β, interleukin [IL]-10, and IL35), expression of co-inhibitory

Table 1

IEI resulting in loss of tolerance and inflammation in the gastrointestinal tract.

Inborn error of immunity	Gene (s) affected	Mechanism	Gastrointestinal phenotype
IPEX syndrome	*FOXP3*	Impaired Treg differentiation	Autoimmune enteropathy: predominantly proximal inflammation with autoantibody formation
IPEX-like syndrome	*IL2RA, STAT5B*		
APECED	*AIRE*	Impaired central tolerance and defective t-Treg development	
CTLA-4 haploinsufficiency	*CTLA4*	Impaired cell-mediated immunoregulation	
LRBA deficiency	*LRBA*		
IL-10 and IL-10R deficiency	*IL10, IL10RA, IL10RB*	Impaired IL-10 signalling axis	VEOIBD: dominantly colonic inflammation with deep ulceration and fistula formation
Chronic granulomatous disease	*CYBB, CYBA, NCF1, NCF2, NCF4*	Defective NADPH oxidase activity, leading to impaired oxidative burst and defective innate microbial elimination	IBD or VEOIBD with high antimicrobial antibody titres

Selected examples of IEI, which result in loss of tolerance and inflammation in the gastrointestinal tract. IPEX immune dysregulation, polyendocrinopathy, enteropathy X-linked syndrome. APECED Autoimmune polyendocrinopathy-candidiasis ectodermal dystrophy; NADPH nicotinamide adenine dinculeotide phosphate.

receptors (CTLA4, ICOS, and TIGIT), and direct cytotoxicity via perforin and granzyme release [21]. Immunoregulatory functions are also performed constitutively by CD4+ FOXP3- type-1 regulatory T (Tr1) cells [19] and can be induced transiently in otherwise inflammatory CD4+ effector T cells [27].

Loss of tolerance due to regulatory T cell defects

Loss of tolerance can occur due to defects in Treg differentiation or effector functions or because mucosal immune homeostasis is disturbed (e.g. by epithelial or innate defects). Monogenic IEI can disrupt these processes, enabling identification of nonredundant tolerance mechanisms in the intestine and assessment of how they relate to IBD (Table 1).

Defective Treg differentiation occurs secondary to loss-of-function variants in *FOXP3* or the IL*2* signalling pathway (IPEX and IPEX-like syndrome). These syndromes cause severe autoimmune enteropathy, often most severe in the small bowel and accompanied by anti-enterocyte autoantibody production [28,29]. Enteropathy in these conditions is evident *in utero*, and FOXP3$^{-/-}$ mice develop intestinal pathology in germ-free conditions, confirming defective tolerance to self-antigen is the dominant pathogenic mechanism in global Treg deficiency [30,31]. However, IL2$^{-/-}$ mice that lack Tregs develop a microbiota-dependent colitis alongside systemic autoimmunity [32], indicating co-existing loss of tolerance to microbiota and autoimmunity. This may be particularly significant in a subgroup of patients with late-onset or atypical IPEX, who can develop IBD-like inflammation [29].

Loss-of-function variants in *AIRE* disrupt central tolerance mechanisms, resulting in autoreactive naïve effector T cells and impaired tTreg development. This results in widespread autoimmunity (APECED syndrome) frequently including autoimmune enteropathy (typically not IBD) [33]. The disease is characterised by autoantibodies against enterochromaffin cells [33] and mucosal-protective cytokines IL17A, IL17F, and IL*22*, which can cause chronic mucosal candidiasis [34].

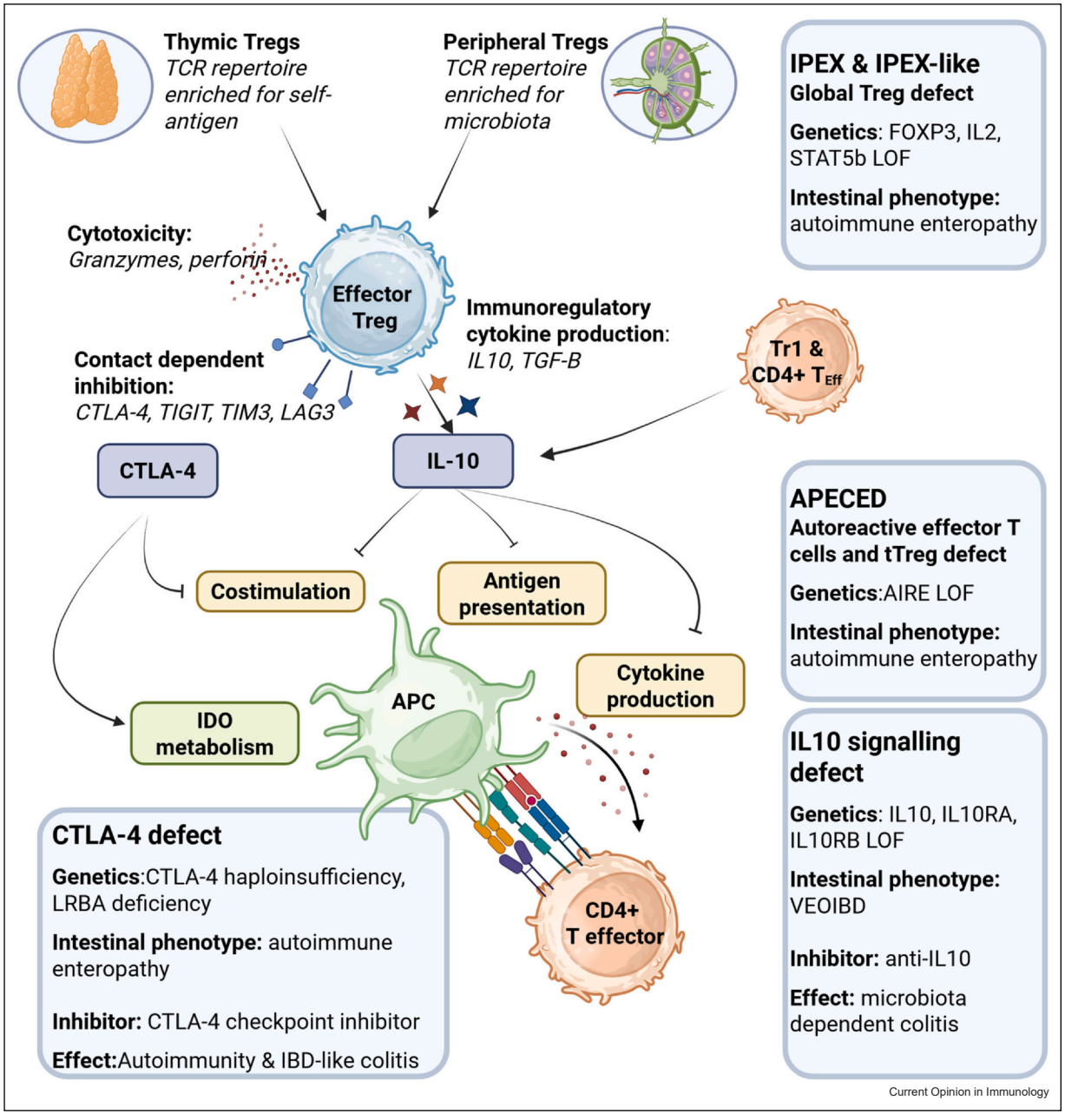
Tregs employ several immunoregulatory effector functions to promote tolerance (Figure 1). The best characterised are direct cell-mediated inhibition by the co-inhibitory receptor CTLA-4 and production of IL-10. CTLA-4 is expressed by Tregs constitutively and conventional T cells following activation. It binds and downregulates CD80 and CD86 on APCs, preventing activation of the co-stimulatory CD28 receptor [19]. CTLA-4 haploinsufficiency or deficiency of *LRBA* (required for CTLA-4 membrane recycling) cause an autoimmune enteropathy resembling IPEX [28]. Inhibition of CTLA-4 in cancer immunotherapy induces autoimmune complications and an IBD-like colitis, which can be modulated by antibiotics and faecal microbial transplant [35]. Similarly, in murine colitis transfer models, blocking CTLA-4 on Tregs abrogates their suppressive ability [36], suggesting Treg CTLA-4 promotes tolerance to the microbiota as well as self-antigen.

Global Treg deficiency and defects in contact-mediated inhibition cause severe autoimmune enteropathy with microbiota-dependent inflammation in some contexts, indicating there is some overlap in mechanisms promoting tolerance to self and microbial antigen (Figure 1). Genome-wide association studies implicate common genetic variants with subtle effects on Treg differentiation and function with increased risk of IBD [37].

Interleukin 10 and anti-inflammatory responses

IL-10 has multiple regulatory effects on innate and adaptive immune cells [3,38]. Biallelic loss-of-function variants in IL-10 or its receptor cause severe and fistulising very-early-onset IBD with very high penetrance and are rarely associated with autoimmunity [39,40]. In

Figure 1



Intestinal Tregs maintain tolerance to self and microbial antigen using overlapping effector mechanisms. Intestinal Tregs are generated in the thymus (tTregs) or mLNs (pTregs) where they acquire a TCR repertoire enriched for self or microbial antigens. They utilise contact-dependent inhibition, cytotoxicity, and immunoregulatory cytokines to promote tolerance in the mucosa. The effects of CTLA-4 and IL-10 on APCs are shown. Monogenic diseases or specific inhibition of IL-10 and CTLA-4 results in inappropriate adaptive responses to self-antigen and/or the microbiota. Global Treg defects and defects in CTLA-4 contact-dependent inhibition favour responses to self-antigens, IL-10 defects strongly predispose to antimicrobial immunity. *Created in BioRender. Taylor, H. (2025)* https://BioRender.com/v97z164.

patients with defective IL-10 signalling, Tregs are present in the intestine and their ability to suppress effector T cell expansion *in vitro* is intact, but there is increased pathogenic Th17 polarisation and abnormal clonal expansion of T and B cells [41]. Disrupting the IL-10 signalling axis in mouse models leads to a severe microbiota-dependent colitis [3,26]. Mice develop a milder phenotype when IL-10 deficiency is restricted to FOXP3+ Tregs, as IL-10 from other CD4+ T cells partially compensates for the Treg defect [42].

Defective epithelial and phagocytic barrier function drives adaptive immune responses

Intestinal monocytes and macrophages are the key IL-10-responsive cells in the intestinal mucosa. They have constitutive high surface expression of IL10R, deficiency of which reproduces the phenotype of global IL10 deficiency [43]. IL10 signalling induces a metabolic shift towards oxidative phosphorylation, associated with reduced production of pro-inflammatory cytokines, reduced antigen presentation, and increased intracellular bactericidal activity [38]. These effects are counterbalanced by microbial stimulation; lipopolysaccharide (LPS) stimulation promotes glycolytic metabolism and pro-inflammatory cytokine production [38], while exposure of normal monocytes to whole bacteria induces a state of acquired IL-10 resistance [44].

Defects in epithelial barrier function and phagocytic function increase microbial exposure, thereby countering the effect of IL10 and inducing loss of tolerance to the microbiota. Consistent with this concept is the finding that patients with chronic granulomatous disease (CGD), a monogenic IBD that disrupts innate intracellular killing of bacteria, have high antimicrobial antibody titres even in the absence of intestinal inflammation [45]. Similarly, IBD patients with variants in *NOD2* or *IRGM* [46], and with NFkB1 haplotypes [47] that impair microbial handling have higher titres of antimicrobial antibodies.

Disrupting the IL10 axis, through genetic defect or by functionally offsetting the effects of IL10 signalling through microbial stimulation, results in loss of tolerance to the microbiota and IBD. Systemic autoimmunity is uncommon in these conditions, suggesting alternative Treg effector functions are sufficient to maintain tolerance to self-antigen.

Section 3: Inflammatory bowel disease–associated mucosal adaptive responses

Loss of tolerance to self and microbial antigens (the 'holobiont self') is an early checkpoint in IBD pathogenesis and can exist in a predisease state. Antimicrobial antibodies and autoantibodies are detected many years before clinical diagnosis of IBD [48–50], and among healthy first-degree relatives antimicrobial antibodies are associated with future disease development [48]. Patients with established IBD have extensive loss of tolerance, as demonstrated by recent multiplexed immunoglobin detection arrays, which identify hundreds of disease-associated antimicrobial and anti-self antibodies [51–53].

The characteristic serological responses of IBD (Supplementary Table 1), many of which were identified several decades ago, are high-titre and prevalent examples of a much larger group of IBD-associated mucosal adaptive responses. This group encompasses T and B cell responses and includes host-protective, bystander, and pathogenic examples. They have potential utility as biomarkers and therapeutic targets.

Pathogenic T cell responses

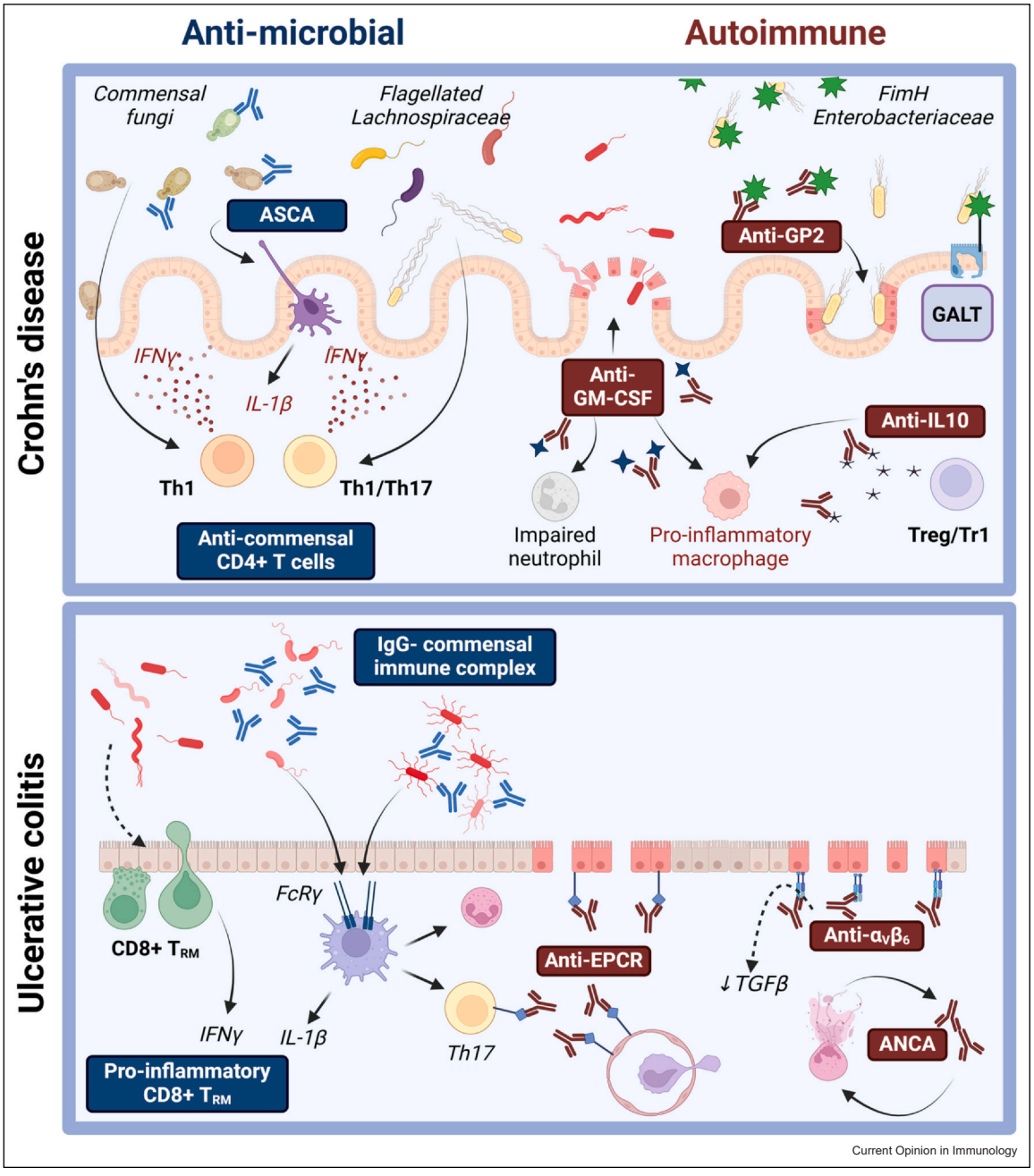
In health, a large, clonally diverse population of commensal-specific CD4+ T cells respond to highly expressed conserved microbial epitopes with predominantly barrier-supporting and immunoregulatory effector functions [18,54]. In IBD, there is complex dysregulation of the CD4+ T cell response [18], with increased Th1 and pro-inflammatory Th17 subtypes and decreased Tregs. The TCR repertoire is clonally expanded and has reduced diversity [55], indicating pathogenic activity against a restricted pool of antigens. Clonal expansions are predominantly private and directed against unidentified antigens [19]. However, several antigen-specific pathogenic CD4+ T cell responses have been identified, particularly against the targets of antimicrobial antibodies (Figure 2).

Anti-*Saccharomyces cerevisiae* antibodies (ASCA) recognise structurally related yeast oligo-mannans on the cell wall of *Saccharomyces cerevisiae*, *Candida albicans*, and other yeasts [56]. ASCA-positive CD patients have strong pro-inflammatory responses to *S. Cerevisiae*, other commensal yeasts, and mould-ripened cheeses, caused by a clonally expanded and chronically activated population of Th1 cytotoxic lymphocytes, cross-reactive against multiple commensal fungi and dietary antigens [57]. Repeated stimulation of these cells by luminal antigen could drive mucosal inflammation.

Seroreactivity against *Lachnospiraceae* flagellins is also common and associated with complex CD [52,58]. A strong antiflagellar Th1 and Th17 response has been identified in CD [58,59]. The cells have a Th1, Th17, or Th17.1 effector memory phenotype [58,59], with a relative depletion of Treg cells [58]. An increased Th17 and reduced Treg response has also been described to *Escherichia coli* Outer-membrane porin C (OmpC) [60] and *Bacteroidetes* SusC antigen [61].

In contrast to CD4+ T cells, there is limited clonal expansion or repertoire alteration of CD8+ T cells in inflamed mucosa [62]. Instead, homologous TCR repertoires between blood and mucosa and expression of cell-trafficking molecules suggest continued circulation of these cells [62]. Nevertheless, cytotoxic CD8+ T cells localise to areas of epithelial damage in IBD [63], and IBD-associated pro-inflammatory transcriptional programmes adopted by CD8+ and unconventional T cells are an active area of interest. In UC, expanded populations of inflammatory CD8+ *EOMES*+ T cells [64] have been described in colonic tissue, while in CD, there is

Figure 2



Pathogenic contributions from antimicrobial and autoimmune IBD-associated mucosal adaptive responses in Crohn's disease (CD) and ulcerative colitis (UC). In ASCA-positive CD patients, glycosylated IgG ASCA binds FcRy receptors on dendritic cells, activating a pro-inflammatory response. IFNy-producing cytotoxic Th1 cells cross-react to *S. cerevisae*, other fungi, and dietary antigens contributing to inflammation. IFNy is also produced by Th1 and Th17 cells responsive to *Lachnospiraceae* flagellins, *E. coli* OmpC antigen, and *Bacteroides* SusC antigen. Anti-GP2 antibodies are induced when bacteria bind to membrane-associated GP2 on M cells overlying GALT and undergo transcytosis, they inhibit luminal antimicrobial function of GP2. Anti-GM-CSF antibodies impair epithelial barrier function and neutrophil phagocytic capacity and alter myeloid retinoic metabolism, which induces pro-inflammatory ILC polarisation. Anti-IL-10 antibodies neutralise the effect of IL-10, leading to pro-inflammatory macrophages. In UC, commensal-IgG complexes bind FcRy receptors, triggering a pro-inflammatory response. Anti-EPCR block anti-inflammatory functions of EPCR on endothelial cells, IECs, and differentiating Th17 cells. Anti $\alpha_v\beta_6$ antibodies disrupt IEC integrity and could reduce TGFβ activation. ANCA antigen is released by neutrophil NET formation, ANCA encourage NET release and delay NET clearance. *Created in BioRender. Taylor, H. (2025)* https://BioRender.com/r56q856.

expansion of a subpopulation of NKT-type II cells with a shared semi-invariant alpha chain [65] and a disease-specific pro-inflammatory and cytotoxic CD103+CD161+ CCR5+CD4+ T_{RM} subset [66].

Pathogenic antibody formation

Abnormal B cell activation in IBD results in clonal expansion, reduced diversity, and a marked increase in IgG+ plasmablasts and plasma cells [63,67]. Anti-commensal IgG antibodies are increased and can have a direct pathogenic effect [52,53,67]. Commensal-IgG complexes engage FcγRIIA on innate immune cells, provoking IL1B release and a Th17 response [68]. An IgG-mediated macrophage activation signature is detectable in ulcerative colitis (UC) anti-TNF non-responders [67], and polymorphisms that reduce the affinity of FcγRIIA for IgG protect against UC [68]. Similarly, a glycosylated ASCA IgG isoform, which is specific for CD and present prediagnosis, triggers a pro-inflammatory response via FcγRIIA binding and increases colitis severity in mice [69].

IBD-associated autoantibodies can cause or exacerbate intestinal inflammation when they impair host-protective responses or neutralise tolerogenic signalling pathways (Figure 2). In Crohn's disease (CD), several autoantibodies are associated with complex ileal disease (Supplementary Table 1) and interfere with barrier or innate antimicrobial function.

Anti-GM-CSF antibodies neutralise GM-CSF, destabilising the homeostatic interaction between IECs and innate immune cells. CD patients with anti-GM-CSF antibodies have altered monocyte retinoic acid production, leading to dysregulated, pro-inflammatory ILC1s [50]. This is accompanied by increased intestinal permeability [70] and reduced neutrophil phagocytic capacity [71]. Variants in the GM-CSF receptor genes confer increased risk of CD with similar complex ileal phenotype [72]; suggesting impaired GM-CSF signalling, by neutralising antibodies or receptor polymorphism has a pathogenic role in CD. Anti-GM-CSF antibodies are also associated with pulmonary alveolar proteinosis (PAP). In PAP, antibodies target non-glycosylated GM-CSF in the respiratory mucosa [50], whereas anti-GM-CSF in CD targets glycosylated epitopes specific to gastrointestinal signalling networks, accounting for the different phenotypes of the two conditions.

Anti-IL10 antibodies disrupt the critical immunoregulatory IL-10 signalling axis and have been identified in an infant with severe VEOIBD. Rapid clinical response was achieved by B cell depletion via a monoclonal antibody Rituximab in addition to conventional therapy, ultimately enabling successful withdrawal of all immunomodulatory treatment 3 years after the anti-IL10 antibody titre became undetectable [73]. Neutralising anti-IL10 antibodies have also been identified in another child with a milder phenotype, suggesting they may be associated with a spectrum of pathology. Although anti-IL10 antibodies may present in IBD patients, there is conflicting data regarding their neutralising capacity [73]. Neutralising anti-IL10R antibodies are used extensively in animal models to induce a strain and microbiota-dependent colitis, supporting a model in which neutralising IL-10 autoantibodies can contribute to inflammation in combination with microbial and other genetic risk factors.

Anti-GP2 antibodies, initially identified as antipancreatic antibodies, impair antimicrobial defence. GP2 is secreted from the pancreas and Brunner's glands and is expressed in a membrane-bound form by M cells in gastrointestinal associated lymphoid tissue (GALT) [74,75]. GP2 binds to FimH-expressing microbiota, for example, CD-associated adherent-invasive-*E coli*, reducing their mucosal colonisation [74,75]. Anti-GP2 antibodies neutralise GP2 in mice, enhancing pathobiont adherence and exacerbating *Salmonella Typhimurium* and dextran sulphate sodium (DSS)-induced colitis [74,76]. On M cells, GP2 facilitates bacterial transcytosis, which can induce anti-GP2 antibody formation [74].

In UC, autoantibodies are associated with disease extent and severity (Supplementary Table 1). Pathogenic mechanisms associated with UC autoantibodies involve disruption of innate, epithelial, and endothelial inflammatory pathways.

Anti-neutrophil cytoplasmic antibodies (ANCA) are classically associated with vasculitis, where the dominant antigens are proteinase-3 and myeloperoxidase. UC-associated ANCA produces an atypical pattern of DNase-1-sensitive perinuclear staining, which is thought to target a DNA-protein complex [77]. The antigen for UC-associated ANCA is released during neutrophil extracellular trap (NET) formation [77]. IBD-associated ANCA reduce NET clearance [78] and augment NET release from TNF-primed neutrophils [79], therefore potentiating a pro-inflammatory response and providing antigen for further autoantigenic stimulation. Though demonstrated in animal models of vasculitis, pathogenicity of ANCA is yet to be shown in colitis models.

Anti-Endothelial Protein C Receptor (EPCR) antibodies target a cell surface receptor for activated protein C. EPCR ligation has anti-inflammatory effects on three distinct cell types, all of which can be blocked by anti-EPCR antibodies. On endothelial cells, EPCR regulates the coagulation cascade and activates an anti-inflammatory response, which inhibits TNF-induced upregulation of cell-adhesion molecules and chemokines and reduces leucocyte adhesion [80]. Endothelial EPCR signalling reduces the severity of DSS-induced colitis in mice, and this protective effect is blocked by anti-EPCR antibodies [80]. On IECs, reduced EPCR signalling is associated

with impaired barrier function and increased severity of DSS colitis [81]. Finally, on differentiating T cells, EPCR ligation downregulates RORyt, IL17A, IL17F, IL21, and IL22; inhibiting Th17 cell polarisation [80]. Anti-EPCR antibodies disinhibited Th17 cell differentiation [80].

Anti-integrin $\alpha_v\beta_6$ antibodies are present in a majority of patients with ulcerative colitis [82] and are present up to 10 years before diagnosis, and titres correlate with adverse disease outcomes [49]. Anti-$\alpha_v\beta_6$ antibodies block the RGD-binding motif of integrin-$\alpha_v\beta_6$, inhibiting enterocyte binding to the extracellular matrix protein fibronectin and reducing epithelial barrier integrity [82]. The integrin $\alpha_v\beta_6$ RGD domain is also known to bind inactive pro-TGF-β in the extracellular matrix, applying force to release the active cytokine [83]. Anti-$\alpha_v\beta_6$ antibodies could disrupt TGF-β activation, though this has not yet been demonstrated. Dysregulated TGF-β signalling is thought to contribute to the severe colitis associated with ITGAV loss-of-function variants, which prevent expression of $\alpha_v\beta_6$ and other α_v integrins [83].

Conclusion

Insights from a holobiont perspective

From a holobiont perspective, symbiotic microbial communities and self-antigen are both components of the 'holobiont self' and must be tolerated to maintain intestinal mucosal homeostasis. To achieve this, the adaptive immune system employs different immunoregulatory effector functions. Contact-dependent suppression by Tregs has a prominent role in preventing responses to self-antigen, while the Treg-monocyte/macrophage IL10 signalling axis is critical in regulation of antimicrobial responses. There is some overlap but limited redundancy between these mechanisms.

Functional defects in these systems lead to loss of tolerance to the 'holobiont self'. This is an important early event in pathogenesis of IBD, often preceding diagnosis. The resulting IBD-associated mucosal adaptive responses include inappropriate T and B cell responses to microbial, self, and environmental antigens.

Therapeutic implications

With increasing use of TCR-clonotyping and multiplexed immunoglobulin detection arrays, the full extent of IBD-associated mucosal adaptive-immune responses is being revealed. These studies identify a combination of shared and patient-specific antimicrobial and anti-self-adaptive responses.

Known IBD-associated mucosal adaptive responses have been assessed as biomarkers for the diagnosis, prognosis, and therapeutic stratification of IBD. The sensitivity and specificity of most is insufficient for routine clinical use (Supplementary Table 1), though interesting disease associations have been noted. Several serological markers for IBD are associated with increased prevalence of extraintestinal manifestations, disease location, and disease severity (Supplementary Table 1). There is insufficient evidence to comprehensively assess other interesting associations, such as response to treatment, or sex-associated differences in antimicrobial immunity and autoimmunity in IBD.

Further profiling of IBD-associated adaptive responses may identify novel clinically useful biomarkers, this may include responses to the intestinal virome, which is altered in IBD and a target of serological responses in UC [52,84]. Alternatively, an increased understanding of patient-specific and shared adaptive immune responses may lead to the development of complex biomarkers, based on patterns of reactivity to groups of microbial, viral, and self-antigens. The recent discovery of anti-integrin $\alpha_v\beta_6$, which has good diagnostic accuracy for UC, and glycosylation-specific IgGs, which have promise as a diagnostic biomarker for CD [69], demonstrates the potential for novel biomarker discovery in IBD.

Beyond their utility as biomarkers, identifying IBD-associated mucosal adaptive responses could facilitate development of highly targeted therapeutics. This could involve the identification and treatment of patients with high-titre of deleterious autoantibodies (e.g. neutralising anti-IL-10 antibodies) with B cell and plasma cell–targeted therapies. Anti-CD20 (rituximab) is not effective in undifferentiated IBD patients [85] but could be in this selected cohort. Alternatively, anti-CD19 antibodies, which provide broader B cell depletion, or anti-CD38 antibodies, which target plasma cells, may provide greater benefit. In addition, patients with autoantibody-mediated disturbance of homeostatic or host-defence mechanisms could receive targeted treatments to restore disrupted pathways. For example, unglycosylated GM-CSF as a potential therapeutic strategy for patients with anti-GM-CSF neutralising antibodies [50].

Targeting antigen-specific T cell responses also holds promise. For example, patients with a significant anti-flagellar CD4+ T cell response could have selective depletion of antiflagellar effector T cells [86] or receive antiflagellar chimeric-antigen-receptor regulatory T cells [87] to reduce pro-inflammatory responses.

In conclusion, inappropriate adaptive immune responses to the holobiont make an important contribution to the pathogenesis of IBD. They are the primary driver of disease in some cases and likely a disease amplifier in a much larger group of patients. The variety and specificity of these IBD-associated mucosal adaptive responses make them challenging to detect and study but also provide an opportunity for biomarker discovery and targeted therapeutic intervention.

Declaration of Competing Interest

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Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.coi.2025.102557.

References and recommended reading

Papers of particular interest, published within the period of review, have been highlighted as:

- • of special interest

1. Broberger O, Perlmann P: **Autoantibodies in human ulcerative colitis**. *J Exp Med* 1959, **110**:657-674.

2. Rudbaek JJ, Agrawal M, Torres J, Mehandru S, Colombel J-F, Jess T: **Deciphering the different phases of preclinical inflammatory bowel disease**. *Nat Rev Gastroenterol Hepatol* 2024, **21**:86-100.

3. Uhlig HH, Powrie F: **Translating Immunology into Therapeutic Concepts for Inflammatory Bowel Disease**. *Annu Rev Immunol* 2018, **36**:755-781.2.

4. Iliev ID, Ananthakrishnan AN, Guo C-J: **Microbiota in inflammatory bowel disease: mechanisms of disease and therapeutic opportunities**. *Nat Rev Microbiol* 2025, https://doi.org/10.1038/s41579-025-01163-0

5. van de Guchte M, Blottière HM, Doré J: **Humans as holobionts: implications for prevention and therapy**. *Microbiome* 2018, **6**:81.

6. Breugelmans T, Oosterlinck B, Arras W, Ceuleers H, Man JD, Hold GL, Winter BYD, Smet A: **The role of mucins in gastrointestinal barrier function during health and disease**. *Lancet Gastroenterol Hepatol* 2022, **7**:455-471.

7. Abraham C, Abreu MT, Turner JR: **Pattern recognition receptor signaling and cytokine networks in microbial defenses and regulation of intestinal barriers: implications for inflammatory bowel disease**. *Gastroenterology* 2022, **162**:1602-1616 e6.

8. Pabst O, Slack E: **IgA and the intestinal microbiota: the importance of being specific**. *Mucosal Immunol* 2020, **13**:12-21.

9. Conrey PE, Denu L, O'Boyle KC, Rozich I, Green J, Maslanka J, Lubin J-B, Duranova T, Haltzman BL, Gianchetti L, *et al.*: **IgA deficiency destabilizes homeostasis toward intestinal microbes and increases systemic immune dysregulation**. *Sci Immunol* 2023, **8**:eade2335.

10. Tang W, Wei Y, Ni Z, Hou K, Luo XM, Wang H: **IgA-mediated control of host-microbial interaction during weaning reaction influences gut inflammation**. *Gut Microbes* 2024, **16**:2323220.

11. Doron I, Mesko M, Li XV, Kusakabe T, Leonardi I, Shaw DG, Fiers WD, Lin W-Y, Bialt-DeCelie M, Román E, *et al.*: **Mycobiota-induced IgA antibodies regulate fungal commensalism in the gut and are dysregulated in Crohn's disease**. *Nat Microbiol* 2021, **6**:1493-1504.

12. Lin YH, Duong HG, Limary AE, Kim ES, Hsu P, Patel SA, Wong WH, Indralingam CS, Liu YC, Yao P, *et al.*: **Small intestine and colon tissue-resident memory CD8+ T cells exhibit molecular heterogeneity and differential dependence on Eomes**. *Immunity* 2023, **56**:207-223 e8.

13. Reina-Campos M, Monell A, Ferry A, Luna V, Cheung KP, Galletti G, Scharping NE, Takehara KK, Quon S, Challita PP, *et al.*: **Tissue-resident memory CD8 T cell diversity is spatiotemporally imprinted**. *Nature* 2025, **639**:483-492, https://doi.org/10.1038/s41586-024-08466-x

14. Godfrey DI, Uldrich AP, McCluskey J, Rossjohn J, Moody DB: **The burgeoning family of unconventional T cells**. *Nat Immunol* 2015, **16**:1114-1123.

15. Rezende RM, Cox LM, Moreira TG, Liu S, Boulenouar S, Dhang F, LeServe DS, Nakagaki BN, Lopes JR, Tatematsu BK, *et al.*: **Gamma-delta T cells modulate the microbiota and fecal micro-RNAs to maintain mucosal tolerance**. *Microbiome* 2023, **11**:32.

16. Tyler CJ, Hoti I, Griffiths DD, Cuff SM, Andrews R, Keisker M, Ahmed R, Hansen HP, Lindsay JO, Stagg AJ, *et al.*: **IL-21 conditions antigen-presenting human γδ T-cells to promote IL-10 expression in naïve and memory CD4+ T-cells**. *Discov Immunol* 2024, **3**:kyae008.

17. du Halgouet A, Bruder K, Peltokangas N, Darbois A, Obwegs D, Salou M, Thimme R, Hofmann M, Lantz O, Sagar: **Multimodal profiling reveals site-specific adaptation and tissue residency hallmarks of γδ T cells across organs in mice**. *Nat Immunol* 2024, **25**:343-356.

18. Gomez-Bris R, Saez A, Herrero-Fernandez B, Rius C, Sanchez-Martinez H, Gonzalez-Granado JM: **CD4 T-cell subsets and the pathophysiology of inflammatory bowel disease**. *Int J Mol Sci* 2023, **24**:2696.

19. Krovi SH, Kuchroo VK: **Activation pathways that drive CD4+ T cells to break tolerance in autoimmune diseases**. *Immunol Rev* 2022, **307**:161-190.

20. Hackstein C-P, Costigan D, Drexhage L, Pearson C, Bullers S, Ilott N, Akther HD, Gu Y, FitzPatrick MEB, Harrison OJ, *et al.*: **A conserved population of MHC II-restricted, innate-like, commensal-reactive T cells in the gut of humans and mice**. *Nat Commun* 2022, **13**:7472.

21. Dikiy S, Rudensky AY: **Principles of regulatory T cell function**. *Immunity* 2023, **56**:240-255.

22. Zegarra-Ruiz DF, Kim DV, Norwood K, Kim M, Wu W-JH, Saldana-Morales FB, Hill AA, Majumdar S, Orozco S, Bell R, *et al.*: **Thymic development of gut-microbiota-specific T cells**. *Nature* 2021, **594**:413-417.

23. • Ramanan D, Chowdhary K, Candéias SM, Sassone-Corsi M, Gelineau A, Mathis D, Benoist C: **Homeostatic, repertoire and transcriptional relationships between colon T regulatory cell subsets**. *Proc Natl Acad Sci* 2023, **120**:e2311566120.
This study analyses Treg transcriptional programmes and TCR repertoires from monocolonised mice and mice with immunoregulatory transcription factor knockouts.

24. Kedmi R, Littman DR: **Antigen-presenting cells as specialized drivers of intestinal T cell functions**. *Immunity* 2024, **57**:2269-2279.

25. Burton OT, Bricard O, Tareen S, Gergelits V, Andrews S, Biggins L, Roca CP, Whyte C, Junius S, Brajic A, *et al.*: **The tissue-resident regulatory T cell pool is shaped by transient multi-tissue migration and a conserved residency program**. *Immunity* 2024, **57**:1586-1602 e10.

26. Gu Y, Bartolomé-Casado R, Xu C, Bertocchi A, Janney A, Heuberger C, Pearson CF, Teichmann SA, Thornton EE, Powrie F: **Immune microniches shape intestinal Treg function**. *Nature* 2024, **628**:854-862.

27. Brockmann L, Tran A, Huang Y, Edwards M, Ronda C, Wang HH, Ivanov II: **Intestinal microbiota-specific Th17 cells possess regulatory properties and suppress effector T cells via c-MAF and IL-10**. *Immunity* 2023, **56**:2719-2735 e7.

28. Bildstein T, Charbit-Henrion F, Azabdaftari A, Cerf-Bensussan N, Uhlig HH: **Cellular and molecular basis of proximal small intestine disorders**. *Nat Rev Gastroenterol Hepatol* 2024, **21**:687-709.

29. Consonni F, Ciullini Mannurita S, Gambineri E: **Atypical presentations of IPEX: expect the unexpected**. *Front Pedia* 2021, **9**, https://doi.org/10.3389/fped.2021.643094

30. Carneiro-Sampaio M, Moreira-Filho CA, Bando SY, Demengeot J, Coutinho A: **Intrauterine IPEX**. *Front Pedia* 2020, **8**:599283.

31. Chervonsky AV: **Microbiota and autoimmunity**. *Cold Spring Harb Perspect Biol* 2013, **5**:a007294.

32. Waidmann M, Bechtold O, Frick J-S, Lehr H-A, Schubert S, Dobrindt U, Loeffler J, Bohn E, Autenrieth IB: **Bacteroides vulgatus protects against *Escherichia coli*-induced colitis in gnotobiotic interleukin-2-deficient mice**. *Gastroenterology* 2003, **125**:162-177.

33. Garelli S, Costa MD, Sabbadin C, Barollo S, Rubin B, Scarpa R, Masiero S, Fierabracci A, Bizzarri C, Crinò A, *et al*.: **Autoimmune polyendocrine syndrome type 1: an Italian survey on 158 patients**. *J Endocrinol Investig* 2021, **44**:2493.

34. Philippot Q, Casanova J-L, Puel A: **Candidiasis in patients with APS-1: low IL-17, high IFN-γ, or both?** *Curr Opin Immunol* 2021, **72**:318-323.

35. Westdorp H, Sweep MWD, Gorris MAJ, Hoentjen F, Boers-Sonderen MJ, van der Post RS, van den Heuvel MM, Piet B, Boleij A, Bloemendal HJ, *et al*.: **Mechanisms of immune checkpoint inhibitor-mediated colitis**. *Front Immunol* 2021, **12**, https://doi.org/10.3389/fimmu.2021.768957

36. Read S, Greenwald R, Izcue A, Robinson N, Mandelbrot D, Francisco L, Sharpe A, Powrie F: **Blockade of CTLA-4 on CD4+CD25+ regulatory T cells abrogates their function in vivo**. *J Immunol (Baltim, Md: 1950)* 2006, **177**:4376.

37. Bossini-Castillo L, Glinos DA, Kunowska N, Golda G, Lamikanra AA, Spitzer M, Soskic B, Cano-Gamez E, Smyth DJ, Cattermole C, *et al*.: **Immune disease variants modulate gene expression in regulatory CD4+ T cells**. *Cell Genom* 2022, **2**.

38. Ouyang W, O'Garra A: **IL-10 family cytokines IL-10 and IL-22: from basic science to clinical translation**. *Immunity* 2019, **50**:871-891.

39. Sharifinejad N, Zaki-Dizaji M, Sepahvandi R, Fayyaz F, dos Santos Vilela MM, ElGhazali G, Abolhassani H, Ochs HD, Azizi G: **The clinical, molecular, and therapeutic features of patients with IL10/IL10R deficiency: a systematic review**. *Clin Exp Immunol* 2022, **208**:281-291.

40. Glocker E-O, Kotlarz D, Boztug K, Gertz EM, Schäffer AA, Noyan F, Perro M, Diestelhorst J, Allroth A, Murugan D, *et al*.: **Inflammatory bowel disease and mutations affecting the interleukin-10 receptor**. *N Engl J Med* 2009, **361**:2033-2045.

41. Werner L, Lee YN, Rechavi E, Lev A, Yerushalmi B, Ling G, Shah N, Uhlig HH, Weiss B, Somech R, *et al*.: **Alterations in T and B cell receptor repertoires patterns in patients with IL10 signaling defects and history of infantile-onset IBD**. *Front Immunol* 2020, **11**:109.

42. Rubtsov YP, Rasmussen JP, Chi EY, Fontenot J, Castelli L, Ye X, Treuting P, Siewe L, Roers A, Henderson WR, *et al*.: **Regulatory T Cell-Derived Interleukin-10 Limits Inflammation at Environmental Interfaces**. *Immunity* 2008, **28**:546-558.

43. Zigmond E, Bernshtein B, Friedlander G, Walker CR, Yona S, Kim K-W, Brenner O, Krauthgamer R, Varol C, Müller W, *et al*.: **Macrophage-restricted interleukin-10 receptor deficiency, but not IL-10 deficiency, causes severe spontaneous colitis**. *Immunity* 2014, **40**:720-733.

44. Aschenbrenner D, Quaranta M, Banerjee S, Ilott N, Jansen J, Steere B, Chen Y-H, Ho S, Cox K, Arancibia-Cárcamo CV, *et al*.: **Deconvolution of monocyte responses in inflammatory bowel disease reveals an IL-1 cytokine network that regulates IL-23 in genetic and acquired IL-10 resistance**. *Gut* 2020, **70**:1023.

45. Yu JE, De Ravin SS, Uzel G, Landers C, Targan S, Malech HL, Holland SM, Cao W, Harpaz N, Mayer L, *et al*.: **High levels of Crohn's disease-associated anti-microbial antibodies are present and independent of colitis in chronic granulomatous disease**. *Clin Immunol* 2011, **138**:14-22.

46. Murdoch TB, Xu W, Stempak JW, Landers C, Targan S, Rotter J, Silverberg MS: **Pattern recognition receptor and autophagy gene variants are associated with development of antimicrobial antibodies in Crohn's disease**. *Inflamm Bowel Dis* 2012, **18**:1743.

47. Takedatsu H, Taylor KD, Mei L, McGovern DPB, Landers CJ, Gonsky R, Cong Y, Vasiliauskas EA, Ippoliti A, Elson CO, *et al*.: **Linkage of CD-related serological phenotypes: NFKB1 haplotypes are associated with anti-CBir1 & ASCA, and show reduced NF-κB activation**. *Gut* 2009, **58**:60-67.

48. Lee S-H, Turpin W, Espin-Garcia O, Raygoza Garay JA, Smith MI, Leibovitzh H, Goethel A, Turner D, Mack D, Deslandres C, *et al*.: **Anti-microbial antibody response is associated with future onset of Crohn's disease independent of biomarkers of altered gut barrier function, subclinical inflammation, and genetic risk**. *Gastroenterology* 2021, **161**:1540-1551.

49. • Livanos AE, Dunn A, Fischer J, Ungaro RC, Turpin W, Lee S-H, Rui S, Del Valle DM, Jougon JJ, Martinez-Delgado G, *et al*.: **Anti-integrin αvβ6 autoantibodies are a novel biomarker that antedate ulcerative colitis**. *Gastroenterology* 2023, **164**:619-629.
This study identifies diagnostic and prognostic association of αvβ6 antibodies in UC using multiple cohorts.

50. • Mortha A, Remark R, Valle DMD, Chuang L-S, Chai Z, Alves I, Azevedo C, Gaifem J, Martin J, Petralia F, *et al*.: **Neutralizing anti-granulocyte macrophage-colony stimulating factor autoantibodies recognize post-translational glycosylations on granulocyte macrophage-colony stimulating factor years before diagnosis and predict complicated Crohn's disease**. *Gastroenterology* 2022, **163**:659-670.
This study identifies anti-glycosylated GM-CSF antibodies, which precede the onset of ileal CD and describes their effect on myeloid and innate lymphoid cells in the mucosa.

51. Jodeleit H, Milchram L, Soldo R, Beikircher G, Schönthaler S, Al-amodi O, Wolf E, Beigel F, Weinhäusel A, Siebeck M, *et al*.: **Autoantibodies as diagnostic markers and potential drivers of inflammation in ulcerative colitis**. *PLoS One* 2020, **15**:e0228615.

52. • Bourgonje AR, Andreu-Sánchez S, Vogl T, Hu S, Vila AV, Gacesa R, Leviatan S, Kurilshikov A, Klompus S, Kalka IN, *et al*.: **Phage-display immunoprecipitation sequencing of the antibody epitope repertoire in inflammatory bowel disease reveals distinct antibody signatures**. *Immunity* 2023, **56**:1393-1409 e6.
This study performs a multiplexed immunoglobulin detection assay on a large cohort of IBD patients and healthy controls.

53. Shome M, Song L, Williams S, Chung Y, Murugan V, Park JG, Faubion W, Pasha SF, Leighton JA, LaBaer J, *et al*.: **Serological profiling of Crohn's disease and ulcerative colitis patients reveals anti-microbial antibody signatures**. *World J Gastroenterol* 2022, **28**:4089-4101.

54. Nagashima K, Zhao A, Atabakhsh K, Bae M, Blum JE, Weakley A, Jain S, Meng X, Cheng AG, Wang M, *et al*.: **Mapping the T cell repertoire to a complex gut bacterial community**. *Nature* 2023, **621**:162-170.

55. Saravanarajan K, Douglas AR, Ismail MS, Omorogbe J, Semenov S, Muphy G, O'Riordan F, McNamara D, Nakagome S: **Genomic profiling of intestinal T-cell receptor repertoires in inflammatory bowel disease**. *Genes Immun* 2020, **21**:109-118.

56. Sendid B, Cornu M, Cordier C, Bouckaert J, Colombel JF, Poulain D: **From ASCA breakthrough in Crohn's disease and *Candida albicans* research to thirty years of investigations about their meaning in human health**. *Autoimmun Rev* 2024, **23**:103486.

57. • Martini GR, Tikhonova E, Rosati E, DeCelie MB, Sievers LK, Tran F, Lessing M, Bergfeld A, Hinz S, Nikolaus S, *et al*.: **Selection of cross-reactive T cells by commensal and food-derived yeasts drives cytotoxic TH1 cell responses in Crohn's disease**. *Nat Med* 2023, **29**:2602-2614.
This study describes a cytotoxic pro-inflammatory CD4+ T cell response to fungal antigens in ASCA seropositive CD patients.

58. Alexander KL, Zhao Q, Reif M, Rosenberg AF, Mannon PJ, Duck LW, Elson CO: **Human microbiota flagellins drive adaptive immune responses in Crohn's disease**. *Gastroenterology* 2021, **161**:522-535 e6.

59. Morgan NN, Duck LW, Wu J, Rujani M, Thomes PG, Elson CO, Mannon PJ: **Crohn's disease patients uniquely contain inflammatory responses to flagellin in a CD4 effector memory subset**. *Inflamm Bowel Dis* 2022, **28**:1893-1903.

60. Uchida AM, Boden EK, James EA, Shows DM, Konecny AJ, Lord JD: ***Escherichia coli*–specific CD4+ T cells have public T-cell receptors and low interleukin 10 production in Crohn's disease**. *Cell Mol Gastroenterol Hepatol* 2020, **10**:507-526.

61. Pedersen TK, Brown EM, Plichta DR, Johansen J, Twardus SW, Delorey TM, Lau H, Vlamakis H, Moon JJ, Xavier RJ, *et al*.: **The CD4+ T cell response to a commensal-derived epitope transitions from a tolerant to an inflammatory state in Crohn's disease**. *Immunity* 2022, **55**:1909-1923 e6.

62. Williams KG, Kongala R, Shows DM, Konecny AJ, Hindmarch DC, Clarke AS, Lord JD: **T cell repertoire homogeneity and blood-gut overlap in patients with inflammatory bowel disease**. *Cell Mol Gastroenterol Hepatol* 2024, **17**:119-130.

63. Thomas T, Friedrich M, Rich-Griffin C, Pohin M, Agarwal D, Pakpoor J, Lee C, Tandon R, Rendek A, Aschenbrenner D, *et al*.: **A longitudinal single-cell atlas of anti-tumour necrosis factor treatment in inflammatory bowel disease**. *Nat Immunol* 2024, **25**:2152-2165.

64. Boland BS, He Z, Tsai MS, Olvera JG, Omilusik KD, Duong HG, Kim ES, Limary AE, Jin W, Milner JJ, *et al*.: **Heterogeneity and clonal relationships of adaptive immune cells in ulcerative colitis revealed by single-cell analyses**. *Sci Immunol* 2020, **5**:eabb4432.

65. Minervina AA, Pogorelyy MV, Paysen S, Luening U, Degenhardt F, Franke A, Thomas PG, Rosati E: **Crohn's-associated invariant T cells (CAITs) recognise small sulfonate molecules on CD1d**. *Gut* 2024, **73**:205-206.

66. Yokoi T, Murakami M, Kihara T, Seno S, Arase M, Wing JB, Søndergaard JN, Kuwahara R, Minagawa T, Oguro-Igashira E: **Identification of a unique subset of tissue-resident memory CD_4^+ T cells in Crohn's disease**. *Proc Natl Acad Sci U S A* 2023, **120**:e2204269120.

67. • Uzzan M, Martin JC, Mesin L, Livanos AE, Castro-Dopico T, Huang R, Petralia F, Magri G, Kumar S, Zhao Q, *et al*.: **Ulcerative colitis is characterized by a plasmablast-skewed humoral response associated with disease activity**. *Nat Med* 2022, **28**:766-779.

This study analyses intestinal B cell changes associated with ulcerative colitis, identifying a shift toward anti-commensal IgG production and an IgG-mediated macrophage activation response.

68. Castro-Dopico T, Dennison TW, Ferdinand JR, Mathews RJ, Fleming A, Clift D, Stewart BJ, Jing C, Strongili K, Labzin LI, *et al*.: **Anti-commensal IgG drives intestinal inflammation and type 17 immunity in ulcerative colitis**. *Immunity* 2019, **50**:1099-1114 e10.

69. • Gaifem J, Rodrigues CS, Petralia F, Alves I, Leite-Gomes E, Cavadas B, Dias AM, Moreira-Barbosa C, Revés J, Laird RM, *et al*.: **A unique serum IgG glycosylation signature predicts development of Crohn's disease and is associated with pathogenic antibodies to mannose glycan**. *Nat Immunol* 2024, **25**:1692-1703.

This study describes glycosylation patterns of IgG ASCA in IBD and their influence on myeloid cells via *FcRγ receptors.*

70. Nylund CM, D'Mello S, Kim M-O, Bonkowski E, Däbritz J, Foell D, Meddings J, Trapnell BC, Denson LA: **Granulocyte macrophage-colony-stimulating factor autoantibodies and increased intestinal permeability in Crohn disease**. *J Pedia Gastroenterol Nutr* 2011, **52**:542-548.

71. Jurickova I, Collins MH, Chalk C, Seese A, Bezold R, Lake K, Allmen D, Frischer JS, Falcone RA, Trapnell BC, *et al*.: **Paediatric Crohn disease patients with stricturing behaviour exhibit ileal granulocyte–macrophage colony-stimulating factor (GM-CSF) autoantibody production and reduced neutrophil bacterial killing and GM-CSF bioactivity**. *Clin Exp Immunol* 2013, **172**:455-465.

72. Denson LA, Jurickova I, Karns R, Shaw KA, Cutler DJ, Okou D, Valencia CA, Dodd A, Mondal K, Aronow BJ, *et al*.: **Genetic and transcriptomic variation linked to neutrophil granulocyte–macrophage colony-stimulating factor signaling in pediatric Crohn's disease**. *Inflamm Bowel Dis* 2019, **25**:547-560.

73. • Griffin H, Ceron-Gutierrez L, Gharahdaghi N, Ebrahimi S, Davies S, Loo PS, Szabo A, Williams E, Mukhopadhyay A, McLoughlin L, *et al*.: **Neutralizing autoantibodies against interleukin-10 in inflammatory bowel disease**. *N Engl J Med* 2024, **391**:434-441.

This study describes pathogenic high-titre anti-IL10 antibodies in a patient with VEOIBD.

74. Kurashima Y, Kigoshi T, Murasaki S, Arai F, Shimada K, Seki N, Kim Y-G, Hase K, Ohno H, Kawano K, *et al*.: **Pancreatic glycoprotein 2 is a first line of defense for mucosal protection in intestinal inflammation**. *Nat Commun* 2021, **12**:1067.

75. Roggenbuck D, Goihl A, Sowa M, Lopens S, Rödiger S, Schierack P, Conrad K, Sommer U, Jöhrens K, Grützmann R, *et al*.: **Human glycoprotein-2 expressed in Brunner glands – a putative autoimmune target and link between Crohn's and coeliac disease**. *Clin Immunol* 2023, **247**:109214.

76. Derer S, Brethack A-K, Pietsch C, Jendrek ST, Nitzsche T, Bokemeyer A, Hov JR, Schäffler H, Bettenworth D, Grassl GA, *et al*.: **Inflammatory bowel disease–associated GP2 autoantibodies inhibit mucosal immune response to adherent-invasive bacteria**. *Inflamm Bowel Dis* 2020, **26**:1856-1868.

77. Drury B, Hardisty G, Gray RD, Ho G: **Neutrophil extracellular traps in inflammatory bowel disease: pathogenic mechanisms and clinical translation**. *Cell Mol Gastroenterol Hepatol* 2021, **12**:321-333.

78. Li T, Wang C, Liu Y, Li B, Zhang W, Wang L, Yu M, Zhao X, Du J, Zhang J, *et al*.: **Neutrophil extracellular traps induce intestinal damage and thrombotic tendency in inflammatory bowel disease**. *J Crohns Colitis* 2020, **14**:240-253.

79. He Z, Si Y, Jiang T, Ma R, Zhang Y, Cao M, Li T, Yao Z, Zhao L, Fang S, *et al*.: **Phosphotidylserine exposure and neutrophil extracellular traps enhance procoagulant activity in patients with inflammatory bowel disease**. *Thromb Haemost* 2016, **115**:738-751.

80. Mutoh T, Shirai T, Ishii T, Shirota Y, Fujishima F, Takahashi F, Kakuta Y, Kanazawa Y, Masamune A, Saiki Y, *et al*.: **Identification of two major autoantigens negatively regulating endothelial activation in *Takayasu arteritis***. *Nat Commun* 2020, **11**:1253.

81. Vetrano S, Ploplis VA, Sala E, Sandoval-Cooper M, Donahue DL, Correale C, Arena V, Spinelli A, Repici A, Malesci A, *et al*.: **Unexpected role of anticoagulant protein C in controlling epithelial barrier integrity and intestinal inflammation**. *Proc Natl Acad Sci USA* 2011, **108**:19830-19835.

82. Kuwada T, Shiokawa M, Kodama Y, Ota S, Kakiuchi N, Nannya Y, Yamazaki H, Yoshida H, Nakamura T, Matsumoto S, *et al*.: **Identification of an anti–integrin αvβ6 autoantibody in patients with ulcerative colitis**. *Gastroenterology* 2021, **160**:2383-2394 e21.

83. Ghasempour S, Warner N, Guan R, Rodari MM, Ivanochko D, Whittaker Hawkins R, Marwaha A, Nowak JK, Liang Y, Mulder DJ, *et al*.: **Human ITGAV variants are associated with immune dysregulation, brain abnormalities, and colitis**. *J Exp Med* 2024, **221**:e20240546.

84. Tun HM, Peng Y, Massimino L, Sin ZY, Parigi TL, Facoetti A, Rahman S, Danese S, Ungaro F: **Gut virome in inflammatory bowel disease and beyond**. *Gut* 2024, **73**:350-360.

85. Calderón–Gómez E, Panés J: **Rituximab in active ulcerative colitis**. *Gastroenterology* 2012, **142**:174-176.

86. Zhao Q, Duck LW, Huang F, Alexander KL, Maynard CL, Mannon PJ, Elson CO: **CD4+ T cell activation and concomitant mTOR metabolic inhibition can ablate microbiota-specific memory cells and prevent colitis**. *Sci Immunol* 2020, **5**:eabc6373.

87. Boardman DA, Wong MQ, Rees WD, Wu D, Himmel ME, Orban PC, Vent-Schmidt J, Zachos NC, Steiner TS, Levings MK: **Flagellin-specific human CAR Tregs for immune regulation in IBD**. *J Autoimmun* 2023, **134**:102961.